

10. Clock Frequency Accuracy Measurement Circuit (CAC)

10.1 Overview

The Clock Frequency Accuracy Measurement Circuit (CAC) counts pulses of the clock to be measured (measurement target clock) within the time generated by the clock selected as the measurement reference (measurement reference clock), and determines the accuracy depending on whether the number of pulses is within the allowable range. When measurement is complete or the number of pulses within the time generated by the measurement reference clock is not within the allowable range, an interrupt request is generated.

[Table 10.1](#) lists the CAC specifications, [Figure 10.1](#) shows the CAC block diagram, and [Table 10.2](#) lists the CAC I/O pin.

Table 10.1 CAC specifications

Parameter	Specifications
Measurement target clocks	Frequency can be measured for: • Main clock oscillator (CACMCLK) • Sub-clock oscillator (CACSCLK) • HOCO clock (CACHCLK) • MOCO clock (CACMOCLK) • LOCO clock (CACLCLK) • Peripheral module clock B (PCLKB)
Measurement reference clocks	Frequency can be referenced to: • External clock input to the CACREF pin • Main clock oscillator (CACMCLK) • Sub-clock oscillator (CACSCLK) • HOCO clock (CACHCLK) • MOCO clock (CACMOCLK) • LOCO clock (CACLCLK) • Peripheral module clock B (PCLKB)
Selectable function	Digital filter
Interrupt sources	• Measurement end • Frequency error • Overflow
Module-stop function	Module-stop state can be set to reduce power consumption
TrustZone Filter	Security and Privilege attribution can be set

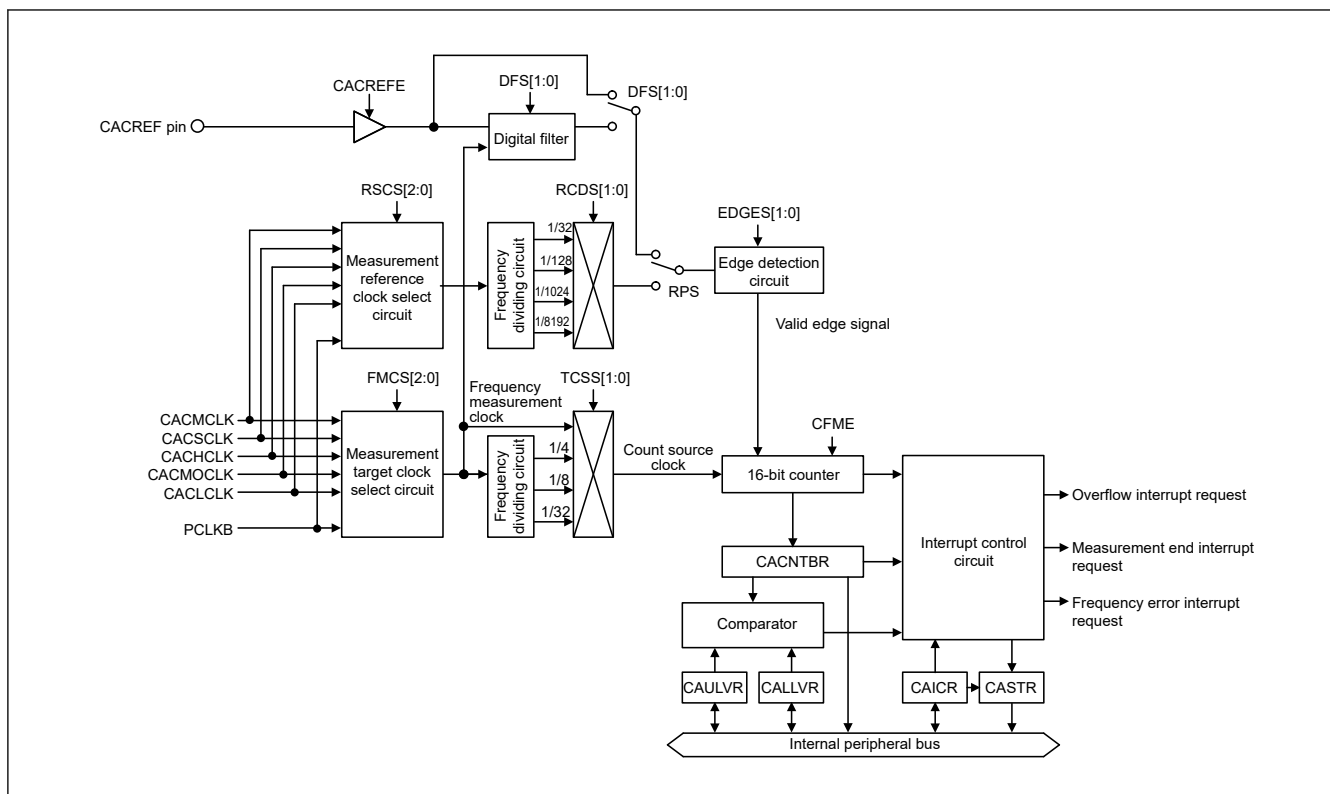


Figure 10.1 CAC block diagram

Table 10.2 CAC I/O pin

Function	Pin name	I/O	Description
CAC	CACREF	Input	Measurement reference clock input pin

10.2 Register Descriptions

10.2.1 CACR0 : CAC Control Register 0

Base address: CAC = 0x4020_2400
CAC_NS = 0x5020_2400

Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CFME

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	CFME	Clock Frequency Measurement Enable 0: Disable 1: Enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

CFME bit (Clock Frequency Measurement Enable)

The CFME bit enables clock frequency measurement. When this bit is rewritten, it takes some time for the change to be reflected in the internal circuit. If this bit is rewritten when the previous value has not been reflected in the internal circuit, the write is ignored. Read the bit to check if the rewrite has been reflected.

10.2.2 CACR1 : CAC Control Register 1

Base address: CAC = 0x4020_2400
CAC_NS = 0x5020_2400

Offset address: 0x01

Bit position:	7	6	5	4	3	2	1	0
Bit field:	EDGES[1:0]		TCSS[1:0]		FMCS[2:0]		CACR EFE	

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	CACREFE	CACREF Pin Input Enable 0: Disable 1: Enable	R/W
3:1	FMCS[2:0]	Measurement Target Clock Select 0 0 0: Main clock oscillator (CACMCLK) 0 0 1: Sub-clock oscillator (CACSCLK) 0 1 0: HOCO clock (CACHCLK) 0 1 1: MOCO clock (CACMOCLK) 1 0 0: LOCO clock (CACLCLK) 1 0 1: Peripheral module clock B (PCLKB) 1 1 0: Setting prohibited 1 1 1: Setting prohibited	R/W
5:4	TCSS[1:0]	Timer Count Clock Source Select 0 0: No division 0 1: × 1/4 clock 1 0: × 1/8 clock 1 1: × 1/32 clock	R/W
7:6	EDGES[1:0]	Valid Edge Select 0 0: Rising edge 0 1: Falling edge 1 0: Both rising and falling edges 1 1: Setting prohibited	R/W

Note: S-TYPE3, P-TYPE3

Note: Set the CACR1 register when the CACR0.CFME bit is 0.

CACREFE bit (CACREF Pin Input Enable)

The CACREFE bit enables the CACREF pin input.

FMCS[2:0] bits (Measurement Target Clock Select)

The FMCS[2:0] bits select the measurement target clock whose frequency is to be measured.

TCSS[1:0] bits (Timer Count Clock Source Select)

The TCSS[1:0] bits select the division ratio of the measurement target clock.

EDGES[1:0] bits (Valid Edge Select)

The EDGES[1:0] bits select the valid edge for the reference signal.

10.2.3 CACR2 : CAC Control Register 2

Base address: CAC = 0x4020_2400
CAC_NS = 0x5020_2400

Offset address: 0x02

Bit position: 7 6 5 4 3 2 1 0

Bit field:	DFS[1:0]	RCDS[1:0]	RSCS[2:0]	RPS
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Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	RPS	Reference Signal Select 0: CACREF pin input 1: Internal clock (internally generated signal)	R/W
3:1	RSCS[2:0]	Measurement Reference Clock Select 0 0 0: Main clock oscillator (CACMCLK) 0 0 1: Sub-clock oscillator (CACSCCLK) 0 1 0: HOCO clock (CACHCLK) 0 1 1: MOCO clock (CACMOCLK) 1 0 0: LOCO clock (CACLCLK) 1 0 1: Peripheral module clock B (PCLKB) 1 1 0: Setting prohibited 1 1 1: Setting prohibited	R/W
5:4	RCDS[1:0]	Measurement Reference Clock Frequency Division Ratio Select 0 0: $\times 1/32$ clock 0 1: $\times 1/128$ clock 1 0: $\times 1/1024$ clock 1 1: $\times 1/8192$ clock	R/W
7:6	DFS[1:0]	Digital Filter Select 0 0: Disable digital filtering 0 1: Use sampling clock for the digital filter as the frequency measuring clock 1 0: Use sampling clock for the digital filter as the frequency measuring clock divided by 4 1 1: Use sampling clock for the digital filter as the frequency measuring clock divided by 16.	R/W

Note: S-TYPE3, P-TYPE3

Note: Set the CACR2 register when the CACR0.CFME bit is 0.

RPS bit (Reference Signal Select)

The RPS bit selects whether to use the CACREF pin input or an internal clock (internally generated signal) as the reference signal.

RSCS[2:0] bits (Measurement Reference Clock Select)

The RSCS[2:0] bits select the reference clock for measurement.

RCDS[1:0] bits (Measurement Reference Clock Frequency Division Ratio Select)

The RCDS[1:0] bits select the frequency-divisor of the reference clock for measurement when an internal reference clock is selected. When RPS = 0 (CACREF pin is used as the reference clock source), the reference clock is not divided.

DFS[1:0] bits (Digital Filter Select)

The DFS[1:0] bits enable or disable the digital filter and selects its sampling clock.

10.2.4 CAICR : CAC Interrupt Control Register

Base address: CAC = 0x4020_2400
CAC_NS = 0x5020_2400

Offset address: 0x03

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	OVFF CL	MEND FCL	FERR FCL	—	OVFIE	MEND IE	FERRI E

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	FERRIE	Frequency Error Interrupt Request Enable 0: Disable 1: Enable	R/W
1	MENDIE	Measurement End Interrupt Request Enable 0: Disable 1: Enable	R/W
2	OVFIE	Overflow Interrupt Request Enable 0: Disable 1: Enable	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	FERRFCL	FERRF Clear 0: No effect 1: The CASTR.FERRF flag is cleared	W
5	MENDFCL	MENDF Clear 0: No effect 1: The CASTR.MENDF flag is cleared	W
6	OVFFCL	OVFF Clear 0: No effect 1: The CASTR.OVFF flag is cleared.	W
7	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

FERRIE bit (Frequency Error Interrupt Request Enable)

The FERRIE bit enables or disables the frequency error interrupt request.

MENDIE bit (Measurement End Interrupt Request Enable)

The MENDIE bit enables or disables the measurement end interrupt request.

OVFIE bit (Overflow Interrupt Request Enable)

The OVFIE bit enables or disables the overflow interrupt request.

FERRFCL bit (FERRF Clear)

Setting the FERRFCL bit to 1 clears the CASTR.FERRF flag.

MENDFCL bit (MENDF Clear)

Setting the MENDFCL bit to 1 clears the CASTR.MENDF flag.

OVFFCL bit (OVFF Clear)

Setting the OVFFCL bit to 1 clears the CASTR.OVFF flag.

10.2.5 CASTR : CAC Status Register

Base address: CAC = 0x4020_2400
CAC_NS = 0x5020_2400

Offset address: 0x04

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	OVFF	MEND F	FERR F

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	FERRF	Frequency Error Flag 0: Clock frequency is within the allowable range 1: Clock frequency has deviated beyond the allowable range (frequency error).	R
1	MENDF	Measurement End Flag 0: Measurement is in progress 1: Measurement ended	R
2	OVFF	Overflow Flag 0: Counter has not overflowed 1: Counter overflowed	R
7:3	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

FERRF flag (Frequency Error Flag)

The FERRF flag indicates a deviation of the clock frequency from the set value (frequency error).

[Setting condition]

- The clock frequency is outside the allowable range defined in the CAULVR and CALLVR registers.

[Clearing condition]

- 1 is written to the FERRFCL bit.

MENDF flag (Measurement End Flag)

The MENDF flag indicates the end of measurement.

[Setting condition]

- Measurement ends.

[Clearing condition]

- 1 is written to the MENDFCL bit.

OVFF flag (Overflow Flag)

The OVFF flag indicates that the counter overflowed.

[Setting condition]

- The counter overflows.

[Clearing condition]

- 1 is written to the CAICR.OVFFCL bit.

10.2.6 CAULVR : CAC Upper-Limit Value Setting Register

Base address: CAC = 0x4020_2400
CAC_NS = 0x5020_2400

Offset address: 0x06

Bit position: 15 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	n/a	The Upper Value of the Allowable Range The CAULVR register is a 16-bit read/write register that specifies the upper value of the allowable range. When the counter value exceeds the value specified in this register, a frequency error is detected. Write to this register when the CACR0.CFME bit is 0. The counter value stored in CACNTBR can vary depending on the difference between the phases of the digital filter and edge-detection circuit, and the signal on the CACREF pin. Ensure that this setting allows an adequate margin.	R/W

Note: S-TYPE3, P-TYPE3

10.2.7 CALLVR : CAC Lower-Limit Value Setting Register

Base address: CAC = 0x4020_2400
CAC_NS = 0x5020_2400

Offset address: 0x08

Bit position: 15 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	n/a	The Lower Value of the Allowable Range The CALLVR register is a 16-bit read/write register that specifies the lower value of the allowable range. When the counter value falls below the value specified in this register, a frequency error is detected. Write to this register when the CACR0.CFME bit is 0. The counter value stored in CACNTBR can vary depending on the difference between the phases of the digital filter and edge-detection circuit, and the signal on the CACREF pin. Ensure that this setting allows an adequate margin.	R/W

Note: S-TYPE3, P-TYPE3

10.2.8 CACNTBR : CAC Counter Buffer Register

Base address: CAC = 0x4020_2400
CAC_NS = 0x5020_2400

Offset address: 0x0A

Bit position: 15 0

Bit field:

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	n/a	The Measurement Result The CACNTBR register is a 16-bit read-only register that stores the measurement result.	R

Note: S-TYPE3, P-TYPE3

10.3 Operation

10.3.1 Measuring Clock Frequency

The CAC measures the clock frequency using the CACREF pin input or an internal clock as a reference. Figure 10.2 shows an operating example of the CAC.

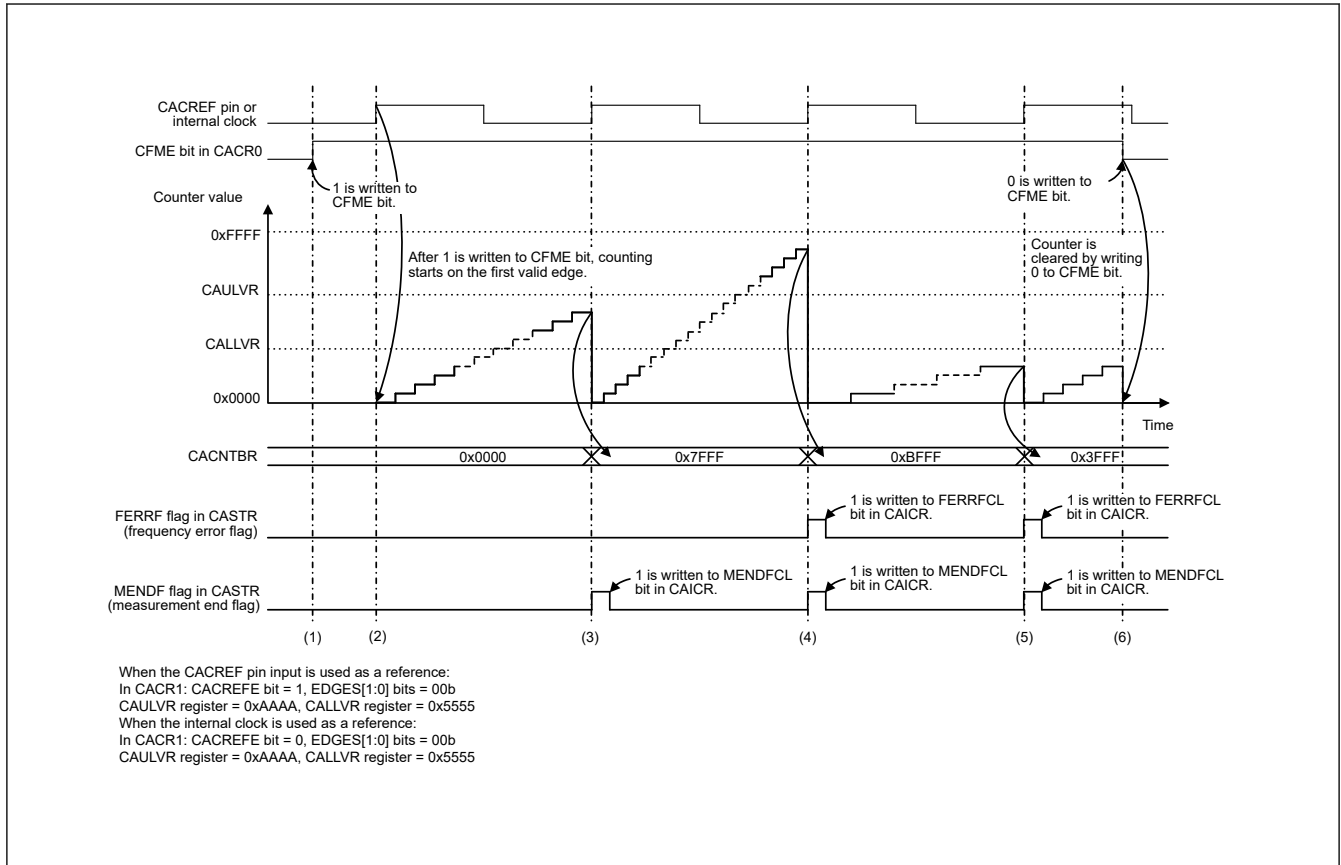


Figure 10.2 CAC operating example

The events in Figure 10.2 are:

1. When the CACREF pin input is used as reference (CACR1.CACREFE = 1), frequency measurement is enabled by writing 1 to the CACR0.CFME bit while the CACR2.RPS bit is set to 0 and the CACR1.CACREFE bit is set to 1. When the internal clock is used as reference (CACR1.CACREFE = 0), frequency measurement is enabled by writing 1 to the CACR0.CFME bit while the CACR2.RPS bit is set to 1.
2. When the CACREF pin input is used as reference, after 1 is written to the CFME bit, the timer starts up-counting if the valid edge selected by the CACR1.EDGES[1:0] bits (rising edge (CACR1.EDGES[1:0] = 00b) in Figure 10.2) is input from the CACREF pin. When the internal clock is used as reference, after 1 is written to the CFME bit, the timer starts up-counting if the valid edge selected by the CACR1.EDGES[1:0] bits (rising edge (CACR1.EDGES[1:0] = 00b) in Figure 10.2) is input based on the clock source selected by the CACR2.RSCS[2:0] bits.
3. When the next valid edge is input, the counter value is transferred to CACNTBR and compared with the values in CAULVR and CALLVR. If both $CACNTBR \leq CAULVR$ and $CACNTBR \geq CALLVR$ are true, only the MENDF flag in CASTR is set to 1, because the clock frequency is correct. If the MENDIE bit in CAICR is 1, a measurement end interrupt is generated.
4. When the next valid edge is input, the counter value is transferred to CACNTBR and compared with the values in CAULVR and CALLVR. If $CACNTBR > CAULVR$, the FERRF flag in CASTR is set to 1, because the clock frequency is erroneous. If the FERRIE bit in CAICR is 1, a frequency error interrupt is generated. The MENDF flag in CASTR is set to 1 at the end of measurement. If the MENDIE bit in CAICR is 1, a measurement end interrupt is generated.
5. When the next valid edge is input, the counter value is transferred to CACNTBR and compared with the values in CAULVR and CALLVR. If $CACNTBR < CALLVR$, the FERRF flag in CASTR is set to 1, because the clock frequency

is erroneous. If the FERRIE bit in CAICR is 1, a frequency error interrupt is generated. The MENDF flag in CASTR is set to 1 at the end of measurement. If the MENDIE bit in CAICR is 1, a measurement end interrupt is generated.

6. When the CFME bit in CACR0 is 1, the counter value is transferred to CACNTBR and compared with the values in CAULVR and CALLVR every time a valid edge is input. Writing 0 to the CFME bit in CACR0 clears the counter and stops up-counting.

10.3.2 Digital Filtering of Signals on CACREF Pin

The CACREF pin has a digital filter, and levels on the CACREF pin are transmitted to the internal circuitry after three consecutive matches in the selected sampling interval. The same level continues to be transmitted internally until the level on the pin has three consecutive matches again. Enabling or disabling of the digital filter and its sampling clock are selectable.

The counter value transferred to CACNTBR might be in error by up to 1 cycle of the sampling clock because of the difference between the phases of the digital filter and the signal input to the CACREF pin. When a frequency dividing clock is selected as a count source clock, the counter value error is obtained using the following formula:

Counter value error = (1 cycle of the count source clock) / (1 cycle of the sampling clock)

10.4 Interrupt Requests

The CAC generates three types of interrupt requests:

- Frequency error interrupt
- Measurement end interrupt
- Overflow interrupt

When an interrupt source is generated, the associated status flag is set to 1. [Table 10.3](#) provides information on the CAC interrupt requests.

Table 10.3 CAC interrupt requests

Interrupt request	Interrupt enable bit	Status flag	Interrupt sources
Frequency error interrupt	CAICR.FERRIE	CASTR.FERRF	The result of comparing CACNTBR with CAULVR and CALLVR is either CACNTBR > CAULVR or CACNTBR < CALLVR
Measurement end interrupt	CAICR.MENDIE	CASTR.MENDF	• Valid edge is input from the CACREF pin or internal clock • Measurement end interrupt does not occur at the first valid edge after writing 1 to the CACR0.CFME bit
Overflow interrupt	CAICR.OVFIE	CASTR.OVFF	Counter overflows

10.5 Usage Notes

10.5.1 Settings for the Module-Stop Function

The Module Stop Control Register C (MSTPCRC) can enable or disable CAC operation. The CAC module is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).